

Question 1

Objective

To determine if there are any downfalls with the current implementation of the lot line labelling algorithm.

Attempts

A QGIS solution to lot line labelling is linked below. It shows the algorithm applied to 1000 Burnaby laneway home parcels.

Evidence

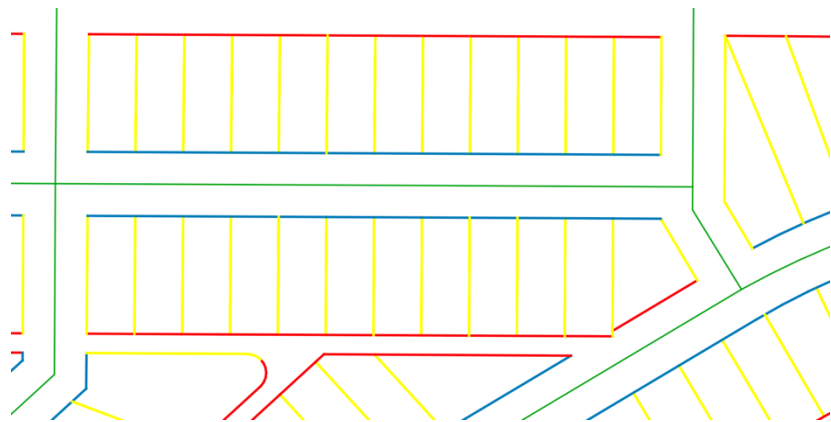
QGIS Solution: https://github.com/ubco-mds-2025-labs/w2025-data599-capstone-projects-green-metrics-technology/tree/main/code/gis_files/burnaby

QGIS Implementation Steps: https://github.com/ubco-mds-2025-labs/w2025-data599-capstone-projects-green-metrics-technology/blob/main/code/gis_files/reports/lot_line_labelling_algorithm_in_qgis.pdf

- Fronts are in blue
- Sides are in yellow
- Rears are in red

Diagnosis

There are many parcel shape variations to explore to gauge the robustness of the algorithm. The algorithm appears robust to us, though there are poor results for some cul-de-sacs.



Proposed Next Steps

Move forward to applying lot line setbacks to define an empty-lot buildable area, followed by building setbacks to get the true buildable area.

Decision Ask

Look at the GIS solution to lot line labelling and explore how the algorithm performs on differently shaped parcels. Do you see any downfalls of the algorithm or implementation that warrant attention before moving forward to defining buildable area?